Supplementary Material

From Offshore Maintenance Records to Fleet Priorities: Evidence Limits and Joint Policies

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This supplement documents the revised analyses. Empirical records, conditional logistics calculations and synthetic policy experiments are reported separately. Tables are generated from the verified numerical outputs; obsolete prescriptions are not retained as sensitivity evidence.

1 Samples, units and observation limits

Table S1: Sample and unit registry. Counts refer to distinct stated populations.

Population or quantity	Value / interpretation
Observation window	1 January 2024–31 March 2026; 821 calendar days
Flight records / fleet exposure	11,926 records / 33,486.6 flight hours
Populated removal records	7,308; not the 32,767 physical worksheet rows
Tracked unscheduled mechanical removals	1,711; 43 lack an aircraft identifier
Panel-linked removals	1,668 before the model-specific exposure exclusions
Full-window NB events	1,499; distinct from rolling training/test samples
Maintenance-task records	57,029
Removal-linked documents / with tasks	1,263 / 1,207
Complete linked-work documents / events	354 / 376
Complete / known linked work	2,043.80 / 33,564.51 labour-hours; different completeness rules
Recorded parts value	USD 56,313,204.83; not actual expenditure
Removal installation intervals	Calendar days; no surviving installed population observed
Empirical logistics time	Calendar hours; 19,704-hour window; assumed 35-day lead time
Synthetic lifetime	Operating days; ageing pauses during downtime

Work is allocated once per linked document, equally across its tracked removals. Complete work requires valid task hours and sign-off dates; unresolved zero-hour administrative orders are excluded. The latest removal/sign-off date is a retrospective information-availability proxy, not a transaction timestamp. Independent safety assessment and observed AOG intervals are unavailable. No pseudo-censored survival sample is constructed.

Table S2: Work completeness by ATA chapter. Zero complete cases are retained.

ATA	Removals	Complete events	Complete (%)
18	2	1	50.00
21	82	23	28.05
22	73	7	9.59
23	40	29	72.50
24	71	33	46.48
25	82	19	23.17

ATA	Removals	Complete events	Complete (%)
26	7	2	28.57
28	8	1	12.50
29	18	0	0.00
31	148	78	52.70
32	282	110	39.01
33	35	13	37.14
34	57	35	61.40
35	2	1	50.00
45	5	2	40.00
53	10	0	0.00
55	5	0	0.00
62	541	7	1.29
63	74	4	5.41
64	91	0	0.00
65	20	0	0.00
67	17	2	11.76
71	7	0	0.00
72	3	0	0.00
73	11	3	27.27
75	9	1	11.11
77	11	5	45.45

2 Rolling forecasts and priority evaluation

Table S3: Mean errors across rolling origins; these are descriptive point comparisons.

Horizon (months)	Method	Origins	MAE	RMSE
3	Cumulative rate	13	1.861290	3.055290
3	NB2	13	1.857304	3.059524
3	Poisson	13	1.849070	3.081009
3	recent 6m rate	13	1.920704	3.321381
6	Cumulative rate	10	2.830874	4.360200
6	NB2	10	2.868181	4.527827
6	Poisson	10	2.845454	4.549057
6	recent 6m rate	10	3.109036	5.036434

Eligibility and estimation use only past observations; counts are conditioned on realised future flight hours. The 46 fitted count models converged. Overlapping origins are not independent replications, and no forecast-interval or significance claim is made.

Table S4: Mean priority coverage over 13 three-month origins; top-three selection with fractional boundary ties.

Method	Observed work (%)	Parts value (%)
Labour weight 0.00	60.0130	57.6304
Labour weight 0.25	58.6440	54.9890
Labour weight 0.50	56.9436	52.2176
Labour weight 0.75	52.0878	48.6569
Labour weight 1.00	41.0403	39.9403
Frequency	60.0130	57.6304

Each origin uses the same training-defined ATA universe across methods, containing 15–18 chapters with both consequences observed. Weights are prespecified preference scenarios. No weight is chosen by test performance; coverage does not measure realised operational savings.

3 Sensitivity to unobserved work

Table S5: Unbounded fixed-selection work envelopes, including known partial-work floors. Bounds are differences in coverage, not confidence limits.

w	Identical selections	Origins permitting positive reversal	Mean infimum	Mean supremum
0.00	13	0	0.000000	0.000000
0.25	12	0	-0.076923	0.000000
0.50	11	0	-0.153846	0.000000
0.75	7	6	-0.461538	0.461538
1.00	0	13	-1.000000	1.000000

The finite-cap grid uses 0, 0.5, 1, 2, 5 and 10 times the training mean complete work per event. All 390 method–origin–cap comparisons fail compatibility with known partial work and are not presented as plausible bounded imputations. The unbounded analysis contains 65 method–origin comparisons. It keeps the original selections fixed and changes the target to eventual work of test removals. Originwise envelopes ignore shared-document constraints and are not a coherent joint imputation across overlapping horizons. They do not account for refitting priorities.

4 Corrected logistics calculations

Table S6: Poisson logistics scenarios with corrected EBO and immediate fill probability.

Scenario	Investment (USD)	EBO	Minimum fill	Wait (h)
S-A	49687.56	25.756272	0.000000	528.097379
S-B	2020082.71	0.840926	0.852501	17.242040
S-C	2462538.52	0.243200	0.950230	4.986482
S-D	199964.69	16.175025	0.000000	331.646926

S-A and S-D impose investment budgets of USD 50,000 and USD 200,000. S-B and S-C impose minimum immediate fill probabilities of 0.85 and 0.95. The 20-part scope assumes quantity per aircraft equal to one and a common 35-day replenishment time. Waiting time is EBO divided by demand rate; it excludes intervention and administrative delays. No aircraft availability or grounded-cost estimate is derived from this ratio. Integer-reference EBO gaps are 0.088% and 0.726% for the budget scenarios. Separate enumeration of 60 three-part synthetic instances finds ten non-optimal heuristic allocations; this is not a bound on empirical fleet performance.

5 Synthetic joint-policy parameters and independent evaluation

Table S7: Base synthetic parameters. None are estimated component-lifetime or AOG parameters of the operator.

Parameter	Value	Unit
assets	8	assets
shape	2.0	dimensionless
scale	180.0	operating days
lead days	20.0	calendar days
corrective days	2.0	calendar days
preventive days	0.5	calendar days

Parameter	Value	Unit
part price	1000.0	USD/part
corrective work cost	500.0	USD/intervention
preventive work cost	150.0	USD/intervention
down cost per day	2000.0	USD/unavailable asset-day
capital rate	0.2	fraction/year
warmup days	365.0	calendar days
measurement days	3650.0	calendar days

The model has one essential position per asset, perfect renewal, unconstrained intervention teams, FIFO backlog and purchased replenishment after every demand. Events preserve on-hand stock plus pipeline minus backlog equal to base stock. Initial inventory investment is separate from annual cost. The selection grid has ten ages (60, 90, 120, 150, 180, 210, 240, 300, 360 and infinity) and 15 stock levels (0–12, 16, 24), with seeds 0–7. Candidates are frozen before 50 paired evaluation seeds 1000–1049. The selected joint policy is age 150 operating days with stock six; corrective stocking uses five. These policies do not have equal initial investment.

Table S8: Joint candidate minus corrective-stock policy: annual cost difference, with individual paired Monte Carlo 95% intervals.

Setting	Difference (USD)	Lower	Upper	Relative (%)
long horizon	-8305.08	-8856.39	-7753.78	-8.268
long warmup	-8579.89	-9433.68	-7726.10	-8.472
no warmup	-7443.78	-8315.28	-6572.28	-7.442
primary	-8233.34	-9045.95	-7420.74	-8.140

Primary warm-up/measurement is 365/3,650 days. Checks use 0/3,650, 1,095/3,650 and 365/7,300 days. Intervals quantify simulation error with fixed inputs, not parameter uncertainty or external validity.

6 Synthetic transfer sensitivities

Table S9: Annual cost contrasts against the scenario-adapted corrective-stock policy. Adapted joint and unchanged base-candidate transfer are distinct comparisons.

Scenario	Policy	Difference (USD)	95% lower	95% upper	Relative (%)
base	Adapted	-8131.05	-9011.10	-7251.01	-8.04
base	Transferred	-8131.05	-9011.10	-7251.01	-8.04
downcost 0	Adapted	-31.20	-60.50	-1.90	-0.13
downcost 0	Transferred	8676.10	8465.12	8887.08	35.80
downcost 200	Adapted	-54.92	-125.87	16.02	-0.15
downcost 200	Transferred	3445.59	3136.99	3754.20	9.71
downcost 20000	Adapted	-141923.97	-150744.24	-133103.70	-18.75
downcost 20000	Transferred	-122936.43	-129840.62	-116032.23	-16.24
lead 0	Adapted	-8291.81	-9175.12	-7408.50	-8.29
lead 0	Transferred	-7091.81	-7975.12	-6208.50	-7.09
lead 5	Adapted	-8178.04	-9063.67	-7292.40	-8.12
lead 5	Transferred	-7783.09	-8670.47	-6895.72	-7.73
lead 60	Adapted	-8885.23	-10139.33	-7631.13	-8.59
lead 60	Transferred	25690.31	21338.53	30042.09	24.84
price 100	Adapted	-15476.07	-16449.75	-14502.38	-18.45
price 100	Transferred	-13455.45	-14218.56	-12692.34	-16.04
price 10000	Adapted	0.00	0.00	0.00	0.00
price 10000	Transferred	46038.04	43569.07	48507.00	16.87

Scenario	Policy	Difference (USD)	95% lower	95% upper	Relative (%)
shape 0.7	Adapted	0.00	0.00	0.00	0.00
shape 0.7	Transferred	61621.97	58511.51	64732.43	60.42
shape 1	Adapted	0.00	0.00	0.00	0.00
shape 1	Transferred	25015.85	22939.01	27092.69	24.66
shape 3	Adapted	-21157.80	-21845.24	-20470.37	-20.97
shape 3	Transferred	-18537.50	-19209.84	-17865.16	-18.37

The twelve scenarios vary one factor at a time: base; Weibull shape 0.7, 1 and 3 with mean lifetime preserved; purchase lead time 0, 5 and 60 days; downtime cost USD 0, 200 and 20,000 per unavailable asset-day; and part price USD 100 and 10,000. Selection uses seeds 0–7, followed by 50 new paired seeds 2000–2049. All scenario candidates are frozen before evaluation. Intervals are individual, without simultaneous multiple-comparison coverage. Adapted policies revert to corrective-only under shapes 0.7 and 1 and high part price. Small cost gains with zero downtime tariff accompany lower availability; the USD 200/day contrast includes zero. No repairable-item workshop, random lead time or interaction between sensitivity factors is represented.

7 Reproduction and anonymisation

The author authorises sharing of operational data after removal of company names, aircraft registrations and other identifying information. Reidentification mappings and identifiable originals are excluded. Anonymisation must preserve dates, exposure, event linkage and missingness needed to reproduce the stated analyses. The current local computational candidate reproduces the synthetic experiments and mathematical tests; this does not establish that the complete revised empirical package is already publicly accessible. Its data release requires verification of the anonymised inputs and their derived results. No superseded safety weighting, pseudocensoring, empirical AOG cost or 81-day replacement prescription is part of this supplement.